COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S3
DATE PERFORMED:	Jul 9, 2026	**DATE SUBMITTED:**	Jul 14, 2026
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ACTIVITY NO. 1.1 **Basic C++ Programming**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A (1.1): Create Code That Follows the Basic C++ Code Structure

```cpp
#include <iostream>

// Create a function that will take two numbers and display the sum.

int sumOfTwoNum(int fNum, int sNum);
void sumOfThreeNum(int fNum, int sNum, int tNum);

int main(){

    std::cout<<"The sum of the two numbers is: " << sumOfTwoNum(6,7)<<std::endl;
    sumOfThreeNum(6,7,8);
}
int sumOfTwoNum(int fNum, int sNum){

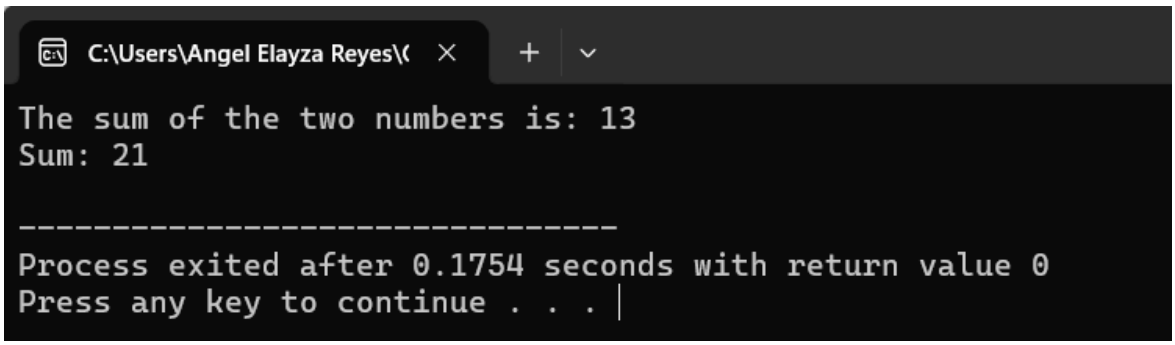
    int sum = fNum + sNum;

        return sum;

}


void sumOfThreeNum(int fNum, int sNum, int tNum){

    std::cout<< "Sum: " << fNum+sNum+tNum << std::endl;
```



ILO A (1.2): Create Code That Follows the Basic C++ Code Structure

ILOA1.cpp [*] ILOA2.cpp

```cpp
#include <iostream>

// Create a function that will return whether variable A is greater than variable B.

bool compareNum(int A, int B);

int main(){

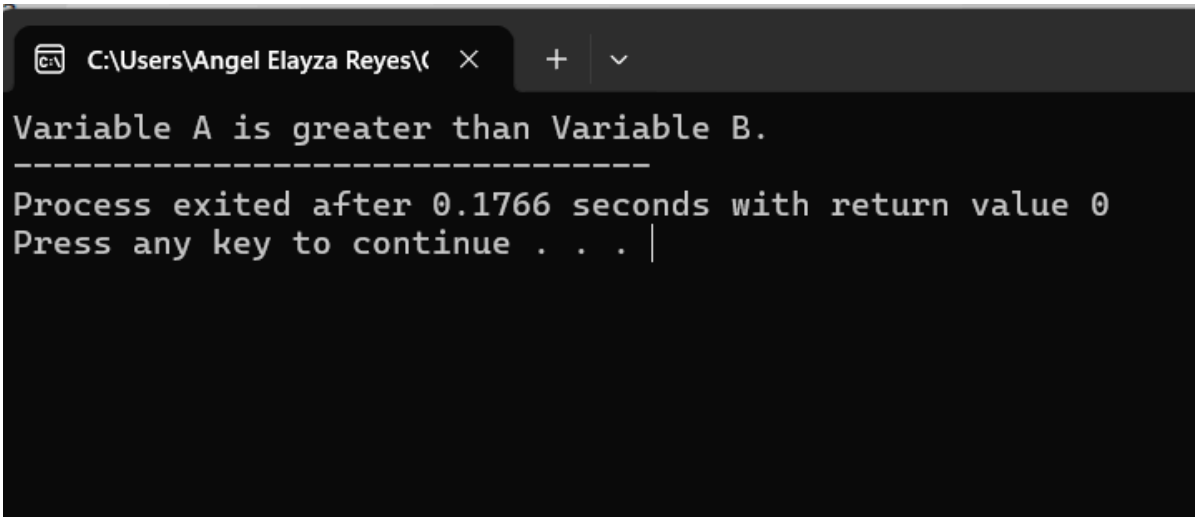
    if(compareNum(100,50)){
        std::cout << "Variable A is greater than Variable B.";
    }
    else{
        std::cout << "Variable A is NOT greater than Variable B.";
    }

    return 0;
}

bool compareNum(int A, int B){

    return A > B;

}
```



ILO A (1.3): Create Code That Follows the Basic C++ Code Structure

ILOA1.cpp | ILOA2.cpp | [*] ILOA3.cpp

```cpp
#include <iostream>

// Create a function that will take two Boolean values and display the result of all
// logical operations then return true if it was a success.

bool logicalOperation(bool A, bool B);

int main(){

    if(logicalOperation(true,false)){
        std::cout << "Logical operations completed successfully.";
    }

    return 0;
}

bool logicalOperation(bool A, bool B){

    std::cout << "A AND B: " << (A && B) << std::endl;
    std::cout << "A OR B: " << (A || B) << std::endl;
    std::cout << "NOT A: " << (!A) << std::endl;
    std::cout << "NOT B: " << (!B) << std::endl;

    return true;

}
```

```
C:\Users\Angel Elayza Reyes\(
A AND B: 0
A OR B: 1
NOT A: 0
NOT B: 1
Logical operations completed successfully.
--------------------------------
Process exited after 0.9021 seconds with return value 0
Press any key to continue . . .
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

ILOA1.cpp | ILOA2.cpp | [*] ILOA3.cpp | [*] ILOB.cpp

```cpp
#include <iostream>
// Use of Triangle Class
// naming convention for class name: Pascalcase

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        //constructor
        Triangle (double A, double B, double C);
        void setAngles(double A, double B, double C);
        //triangle validation
        const bool validateTriangle();
};

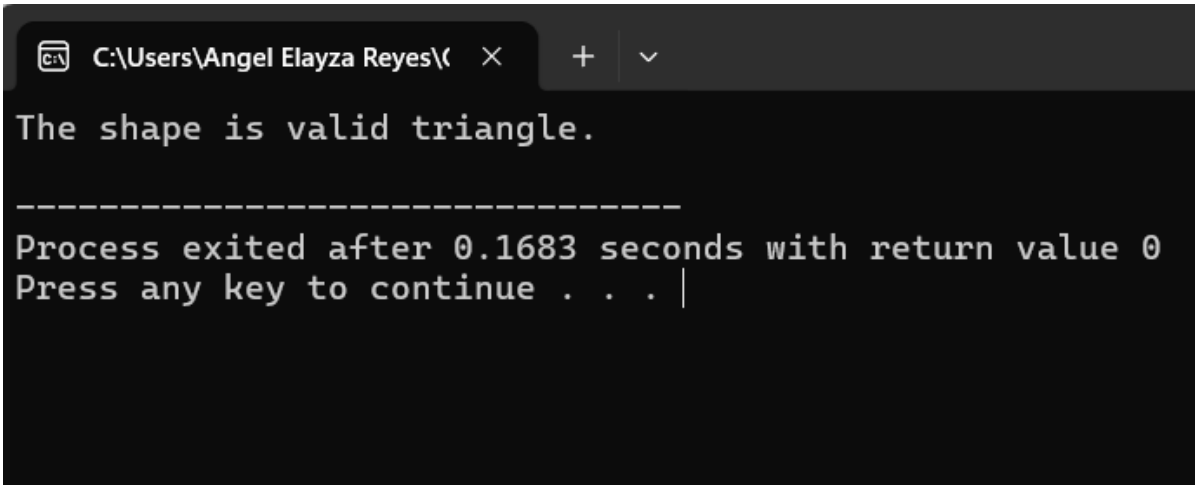
int main(){
    // driver code
    Triangle set(40,30,110);
    if (set.validateTriangle()){
        std::cout<< "The shape is valid triangle. \n";
    }else{
        std::cout<< "The shape is NOT a valid triangle \n";
    }
}
```

```cpp


Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle <=180);
}
```



SECTIONS	ANSWERS
Header File Declaration Section	This is where I include #include <iostream> so my programs can use basic inputs and outputs.
Global Declaration Section	I leave this section blank because I don't use global variables; I pass my variables and parameters directly into my functions instead.
Class Declaration and Method Definition Section	This is where I define my Triangle class from ILOB, setting up its private member variables and the public constructor.
Main Function	This is my int main() block where I create my variables, instantiate the Triangle object, call my functions, and print out the results.
Method Definition	This is where I write the actual logic for all my functions—including the sum and boolean functions from ILOA1, ILOA2, and ILOA3, and the angle validation logic for my Triangle class in ILOB.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create a C++ program to swap the two numbers in different variables.

ILOA1.cpp | ILOA2.cpp | ILOA3.cpp | ILOB.cpp | SUPP1.cpp | SUPP2.cpp | SUPP3.cpp | SUPP4.cpp

```cpp
#include <iostream>
#include <iomanip>

// Swap two numbers using references
void swapNum(int &A, int &B);

int main() {
    int A = 10;
    int B = 20;

    std::cout << "=========================================\n";
    std::cout << "          VARIABLE SWAPPING TEST         \n";
    std::cout << "=========================================\n\n";

    std::cout << " [BEFORE SWAP]\n";
    std::cout << " ------------------------\n";
    std::cout << "  Variable A: " << std::setw(5) << A << "\n";
    std::cout << "  Variable B: " << std::setw(5) << B << "\n\n";

    swapNum(A, B);

    std::cout << " [AFTER SWAP]\n";
    std::cout << " ------------------------\n";
    std::cout << "  Variable A: " << std::setw(5) << A << "\n";
    std::cout << "  Variable B: " << std::setw(5) << B << "\n";
    std::cout << "=========================================\n";

    return 0;
}
void swapNum(int &A, int &B) {
    int temp = A;
    A = B;
    B = temp;
}
```

```
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=========================================
          VARIABLE SWAPPING TEST
=========================================

 [BEFORE SWAP]
 ------------------------
  Variable A:     10
  Variable B:     20

 [AFTER SWAP]
 ------------------------
  Variable A:     20
  Variable B:     10
=========================================

--------------------------------
Process exited after 0.6664 seconds with return value 0
Press any key to continue . . .
```

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2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include <iostream>

// convert temperature of Kelvin to Fahrenheit

double kelvinToFahrenheit(double kelvin);

int main(){

    double kelvin;

    std::cout << "~~~~~ Temperature Converter ~~~~~" << std::endl;
    std::cout << "==================================" << std::endl;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    std::cout << "------------------------------" <<std::endl;
    std::cout << "Temperature in Fahrenheit: "
              << kelvinToFahrenheit(kelvin) << std::endl;

    return 0;
}

double kelvinToFahrenheit(double kelvin){

    return (kelvin - 273.15) * 9 / 5 + 32;

}
```

```
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~~~~~ Temperature Converter ~~~~~
==================================
Enter temperature in Kelvin: 734
--------------------------------
Temperature in Fahrenheit: 861.53

--------------------------------
Process exited after 2.872 seconds with return value 0
Press any key to continue . . .
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include <iostream>
#include <cmath>

// Calculate the distance between two points

double distance(double x1, double y1, double x2, double y2);

int main(){

    double x1, y1, x2, y2;

    std::cout << "====================" << std::endl;
    std::cout << "DISTANCE CALCULATOR" << std::endl;
    std::cout << "====================" << std::endl;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    std::cout << "====================" << std::endl;
    std::cout << "Distance: "
              << distance(x1,y1,x2,y2) << std::endl;

    return 0;
}

double distance(double x1, double y1, double x2, double y2){

    return sqrt(pow(x2-x1,2)+pow(y2-y1,2));

}
```

```
C:\Users\Angel Elayza Reyes\(
====================
DISTANCE CALCULATOR
====================
Enter x1: 12
Enter y1: 14
Enter x2: 24
Enter y2: 18
====================
Distance: 12.6491

--------------------------------
Process exited after 19.67 seconds with return value 0
Press any key to continue . . .
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>

class Triangle{

private:
    double totalAngle, angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    // Constructor
    Triangle(double A, double B, double C,
             double sA, double sB, double sC);

    void setAngles(double A, double B, double C);

    // Triangle validation
    bool validateTriangle();

    // a. Function to compute for the area of a triangle
    double areaTriangle();

    // b. Function to compute for the perimeter of a triangle
    double perimeterTriangle();

    // c. Function that determines whether the triangle is acute-angled, obtuse-angled, or others
    void triangleType();

};

int main(){

    double angleA, angleB, angleC;
    double sideA, sideB, sideC;

    std::cout << "========== TRIANGLE INFORMATION ==========\n";

    std::cout << "Enter Angle A: ";
    std::cin >> angleA;

    std::cout << "Enter Angle B: ";
    std::cin >> angleB;

    std::cout << "Enter Angle C: ";
    std::cin >> angleC;

    std::cout << "\nEnter Side A: ";
    std::cin >> sideA;

    std::cout << "Enter Side B: ";
    std::cin >> sideB;

    std::cout << "Enter Side C: ";
    std::cin >> sideC;

    Triangle set1(angleA, angleB, angleC, sideA, sideB, sideC);

    std::cout << "\n========== RESULT ==========\n";

    if(set1.validateTriangle()){
```

```
61
62          std::cout << "Triangle Validation : Valid Triangle\n";
63
64          // a. Compute the area of the triangle
65          std::cout << "Area                : "
66                    << set1.areaTriangle() << std::endl;
67
68          // b. Compute the perimeter of the triangle
69          std::cout << "Perimeter           : "
70                    << set1.perimeterTriangle() << std::endl;
71
72          // c. Determine the type of triangle
73          std::cout << "Triangle Type       : ";
74          set1.triangleType();
75          std::cout << std::endl;
76
77      }
78      else{
79
80          std::cout << "Triangle Validation : NOT a Valid Triangle\n";
81          std::cout << "Area                : Cannot be computed\n";
82          std::cout << "Perimeter           : Cannot be computed\n";
83          std::cout << "Triangle Type       : Cannot be determined\n";
84
85      }
86
87      return 0;
88  }
89
90  Triangle::Triangle(double A, double B, double C,
91                     double sA, double sB, double sC){
92
93      angleA = A;
94      angleB = B;
95      angleC = C;
96
97      sideA = sA;
98      sideB = sB;
99      sideC = sC;
100
101     totalAngle = A + B + C;
102
103 }
104
105 void Triangle::setAngles(double A, double B, double C){
106
107     angleA = A;
108     angleB = B;
109     angleC = C;
110
111     totalAngle = A + B + C;
112
113 }
```

```cpp

bool Triangle::validateTriangle(){

    return (totalAngle == 180 &&
            angleA > 0 &&
            angleB > 0 &&
            angleC > 0);

}

// a. Function to compute for the area of a triangle
double Triangle::areaTriangle(){

    double s = (sideA + sideB + sideC) / 2;

    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));

}

// b. Function to compute for the perimeter of a triangle
double Triangle::perimeterTriangle(){

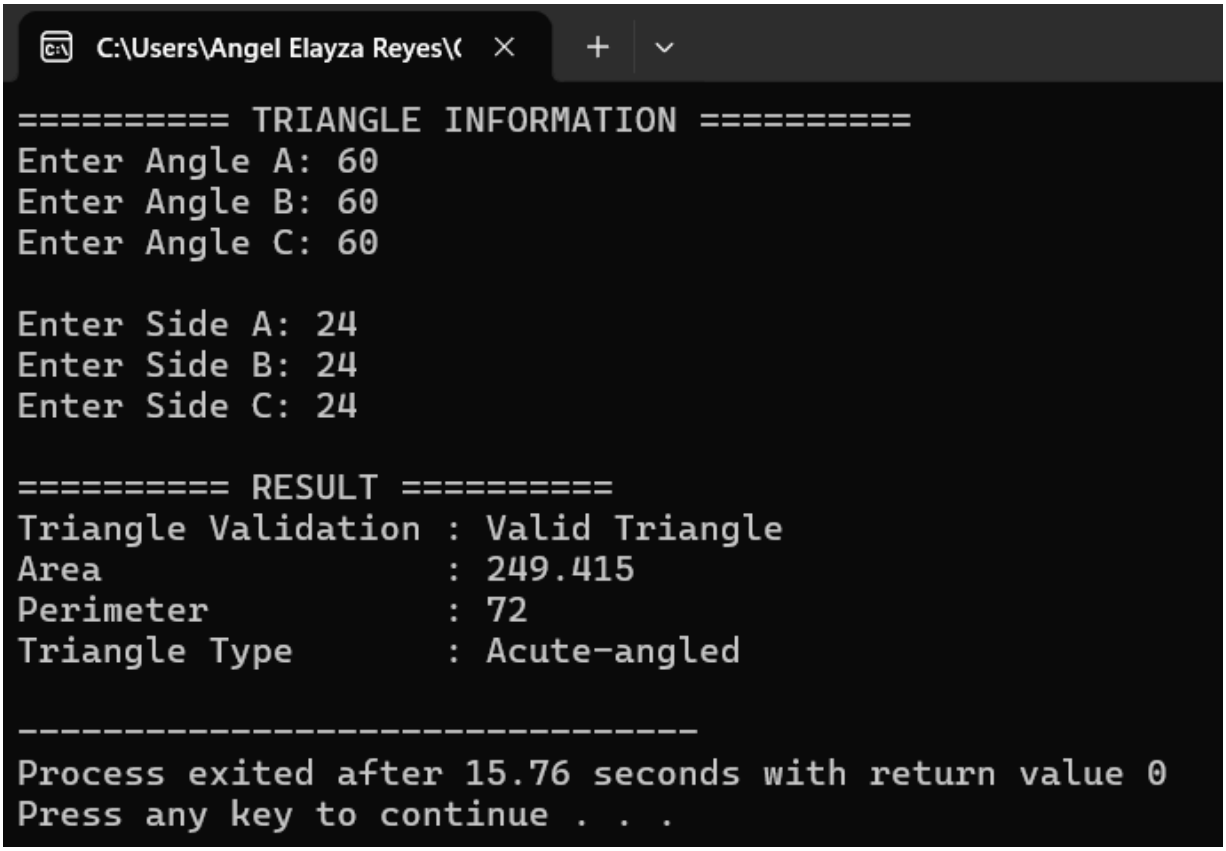
    return sideA + sideB + sideC;

}

// c. Function that determines whether the triangle is acute-angled, obtuse-angled, or others
void Triangle::triangleType(){

    if(angleA < 90 && angleB < 90 && angleC < 90){
        std::cout << "Acute-angled";
    }
    else if(angleA > 90 || angleB > 90 || angleC > 90){
        std::cout << "Obtuse-angled";
    }
    else{
        std::cout << "Others";
    }

}
```



GENERAL CODE ANALYSIS & CONCLUSION::

Going through these activities and tasks paved the way for me to relearn the foundational functional programming concepts and data validation topics. By creating my variable-swapping program, I was able to utilize the pass-by-reference logic to directly manipulate values in memory by swapping two numbers using a temporary variable. My work on the distance equation, moreover, uses a built-in function under the main block, and uses the standard forward declaration process to smoothly run the calculation. Furthermore, designing the Triangle class allowed me to implement basic Object-Oriented Programming principles, where I encountered encapsulated properties such as angles and sides as private members and evaluated them using a separate validateTriangle() function.

While I succeeded in building fully functional code, I have identified areas for improvement on my own current programming habits. I can enhance my program's efficiency through small changes in details like replacing "std::endl;" with "\n" to avoid forcing a slow, unnecessary memory buffer that leads to program's delay and failure. In addition, I also learned that it is necessary to completely understand the given instructions and validation rules such as ensuring the input angles sum up to exactly 180—before going through the coding process in order for the user to use your program more freely and in accuracy. Moving forward, these activities made me realize that I still need to practice my overall coding skills for a more clearer output.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: ChatGPT and Grammarly

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code error explanation, formatting, code assistance, and grammatical correction.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: The AI only paves the way for me to understand concepts and errors that I have made.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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